QP CODE:25900692

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

THIRD SEMESTER MGU-UGP (HONOURS)

REGULAR EXAMINATION OCTOBER 2025

Third Semester

Discipline Specific Core Course : MG3DSCZIM201

BACTERIAL PHYSIOLOGY AND METABOLISM

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall Outcome attainability

Part A

Fill in the blanks

Answer all question

Each question carries 1 marks

1……..use light as their energy source [K]/ [C01]

Answer :Photoautotrophs

2. Vancomycin inhibits bacterial…….. synthesis (K)/(C04)

Answer : Cellwall

3. Anaerobic respiration produces less…….. compared to aerobic respiration [K]/[CO5]

Answer : ATP

4…….. is mainly obtained by plants from the atmosphere in the form of carbon dioxide [K]/[C01]

Answer : Carbon

5. UV radiation causes…….. damage by forming thymine dimers [U]/[CO3]

Answer : DNA

6. Differential media allows microorganisms to be differentiated based on their…….. ……characteristic such as sugar fermentation or enzyme production. [K]/[CO2]

Answer : Biochemical

7. Continous culture are useful in studying ……..under controlled conditions (K)/[CO3]

Answer : Microbial growth

8. In oxidative phosphorylation ATP is produced by the……..[U]/[CO5]

Answer : Chemiosmosis

9. Glycolysis is a metabolic pathway that converts……. into pyruvate

Answer : Glucose

10. The dilution method using broth in microplates is called the……. method [U]/[CO4)

Answer: Microdilution

[1x10=10]

Part –B

Short Questions

Answer any **10** question

Each question carries **2** marks

11. Explain the role of phosphoenolpyruvate in the phosphotransferase system.[K]/ [C01]

Phosphoenolpyruvate (PEP) is the initial phosphate donor in the phosphotransferase system (PTS), providing the energy for the transport and phosphorylation of sugars as they enter the bacterial cell.

12. What are the medical applications of silver compounds? [**U]/ [C03]**

Silver compounds are used in medical applications such as topical creams for burn victims to prevent infection, as coatings on medical devices like catheters, and in wound dressings for their broad-spectrum antimicrobial properties.

13. What is the biological significance of the ED pathway? [U]/ [C05]

The Entner-Doudoroff (ED) pathway is a catabolic pathway in certain bacteria that allows them to metabolize glucose into pyruvate and glyceraldehyde-3-phosphate, serving as an alternative to glycolysis for energy production

14. Explain the mechanism of action of beta-lactam antibiotics. [U]/ [C04]

Beta-lactam antibiotics work by inhibiting the synthesis of the bacterial cell wall. They bind to penicillin-binding proteins (PBPs), which are enzymes responsible for cross-linking peptidoglycan strands, leading to cell lysis and death.

15. What is lactic acid fermentation ? Give examples. [K]/ [C05]

Lactic acid fermentation is a metabolic process where glucose is converted into cellular energy and lactic acid. Examples include the production of yogurt and cheese by Lactobacillus species, and the process in muscle cells during strenuous exercise.

16. What are the benefits of using the spread plate technique? [K]/ [C02]

The spread plate technique is used to evenly distribute a small volume of liquid culture onto an agar plate to obtain isolated colonies. This method allows for the quantification of viable microbial cells in a sample.

17 How does a turbidostat differ from a chemostat? [K]/ [C03]

A turbidostat maintains a constant cell density (turbidity) by adjusting the flow rate of fresh medium, while a chemostat maintains a constant growth rate by keeping the flow rate and nutrient concentration constant.

18. What is the primary function of the TCA Cycle? [U]/ [C05]

The primary function of the Tricarboxylic Acid (TCA) cycle, or Krebs cycle, is to generate energy carriers (ATP, NADH, and FADH2FADH sub 2 FADH2) through the oxidation of acetyl-CoA derived from carbohydrates, fats, and proteins.

19. Define active transport. [K]/ [C01]

Active transport is the movement of molecules or ions across a cell membrane from a lower to a higher concentration region, a process that requires cellular energy, typically in the form of ATP, and specific transport proteins.

20. How does coenzyme Q facilitate electron transfer in the ETS process? [U]/ [C05]

Coenzyme Q (Ubiquinone) is a mobile electron carrier within the electron transport system (ETS) that accepts electrons from complexes I and II and transfers them to complex III, simultaneously pumping protons across the membrane.

21. Why are cultural characteristics important in research and quality control? [U]/ [C02]

Cultural characteristics, such as colony morphology, growth rate, and media requirements, are crucial for the identification, differentiation, and quality control of microorganisms in a laboratory setting.

22. What are the limitations of using low temperature as a microbial control method? [K]/ [C03]

Low temperature methods like refrigeration and freezing inhibit microbial growth and reproduction but are generally not bactericidal; the microorganisms remain viable and can resume growth once the temperature increases.

[2x10=20]

Part – C

Short Essay Question

Answer **5** Questions

Each Question carries **4** Marks

23. Describe the regulation of the HMP pathway and the role of glucose-6-phosphate dehydrogenase deficiency. [U]/ [C05]

(Role of NADP and NADPH -2 Marks , role of Glucose -6 Phosphate- 2 Marks)

The primary regulator of the HMP pathway is the cellular concentration of NADP+NADP raised to the positive power NADP+ .The HMP pathway is primarily regulated by the availability of NADP+ as the first enzyme, glucose-6-phosphate dehydrogenase (G6PD), is allosterically stimulated by high levels of NADP+. G6PD deficiency reduces the production of NADPH, which is crucial for maintaining reduced glutathione levels in red blood cells to neutralize reactive oxygen species, thus deficiency leads to hemolytic anemia when exposed to oxidative stress.

24.Describe the importance of selecting the appropriate media components for the growth and isolation of specific microorganism. [U]/ [C02] Liberal Valuation

(Define growth media- 1 Marks , Any 3 different types of media -3 Marks)

Selecting the appropriate media components is crucial for the successful growth and isolation of specific microorganisms. Selective media contain inhibitors that suppress the growth of unwanted microorganisms while allowing the target organism to thrive.

Differential media include specific substrates or indicators that help distinguish the target organism from others based on a visible change, such as color or pH.

Enrichment media use specific nutrients to favor the growth of a particular microorganism present in small numbers within a mixed population.

The correct media ensures the isolation of a pure culture, which is essential for accurate study, identification, and characterization of the specific microorganism.

25. Explain the viable plate count method with steps and its significance in microbiology. [K]/ [C03] (Significance-1Marks, Any 3 viable plate count method-3 Marks)

The viable plate count method is a technique used to estimate the number of living (viable) microorganisms in a sample.

Step 1: Serial Dilution

A series of dilutions is performed on the original sample to reduce the microbial concentration to a countable range (typically 30-300 colonies per plate).

Step 2: Plating

A known volume (e.g., 0.1 mL or 1 mL) from the selected dilutions is transferred to sterile petri dishes.Molten agar is added and mixed with the sample (pour plate method) or the sample is spread on the surface of a pre-solidified agar plate (spread plate method).

Step 3: Incubation

The plates are incubated under appropriate temperature and atmospheric conditions to allow the viable cells to grow and form visible colonies.

Step 4: Counting and Calculation

Colonies are counted on plates with a countable number. The count is then used to calculate the original concentration of viable cells in the sample, expressed as colony-forming units per milliliter

26. Explain the chemical types of media. [U]/ [C02] (Chemically Defined Media -1 Mark, Complex Media with example -2 Mark, Synthetic Media -1Marks)

Microbiological culture media can be chemically classified into two main types: chemically defined and complex media.

Chemically Defined Media: It is used for metabolic studies

Complex Media: It is rich in nutrients and are used to grow a wide variety of non-fastidious microorganisms.

Synthetic Media: It is used growth conditions for specific, sometimes fastidious, organisms.

27. Discuss the factors that contribute to the onset of the stationary phase in a growth curve.

[U]/ [C03] (Explain any 4 factors 1 mark each)

The stationary phase in a microbial growth curve begins when the rate of cell growth equals the rate of cell death, resulting in a plateau in the number of viable cells.

Nutrient Depletion , Waste Accumulation , Oxygen Limitation and pH

28. Describe cryopreservation and its advantages in microbial preservation. [U]/ [C02] (Definition =1 mark , Advantage=1 mark, Process = 2 Mark)

Cryopreservation is a process of preserving biologically viable materials, such as microbial cultures, by cooling them to very low temperatures.

Long-term Viability: It allows for the preservation of microorganisms for extended periods (decades) with minimal loss of viability.

Genetic Stability: The extremely low temperature essentially halts all metabolic activity and enzymatic reactions, thus preventing genetic drift, mutations, or changes in characteristics that can occur with repeated subculturing.

Contamination Prevention: Once properly frozen and stored, the risk of contamination from external sources is virtually eliminated.

Preservation of Specific Strains: It is a reliable method for maintaining valuable or rare microbial strains for future research, industrial use, or reference purposes.

29. Discuss the involvement of ATP in biosynthetic reaction. [K]/ [C05] (Role of aerobic process 2mark, Role of anaerobic process 2 mark) Liberal Valuation

ATP provides the necessary energy for endergonic biosynthetic reactions. Biosynthetic (anabolic) reactions create complex molecules from simpler ones and require an input of energy. ATP hydrolysis to ADP and Pi releases a significant amount of free energy which is coupled to these endergonic processes, driving the synthesis of macromolecules like proteins, nucleic acids, and lipids. This coupling often involves the transfer of a phosphate group from ATP to a substrate, activating it for the subsequent reaction.

[4x5=20]